Predictive Analysis of TV Series Success and Audience Engagement

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MOtivation & Overview:

I have been watching many TV series since I was a kid. Sometimes, I didn’t understand why a show I really liked was suddenly ending or getting cancelled. This project comes from that personal curiosity. I want to move beyond just being a viewer and use data science to understand why some shows continue for years while others stop.

By following the data science pipeline, I want to see if a show's success depends more on high IMDb ratings (critical acclaim) or on how much "buzz" and search interest it creates on the internet.

Data Sources & Collection Plan:

My primary data will be obtained from the IMDb Non-Commercial Datasets, specifically the title.basics and title.ratings files. I will download these files in TSV format and process them using the Python Pandas library. To enrich this primary data, I will use the **pytrends** library in Python to collect **Google Trends** data.

Data Characteristics:

The final dataset will focus on approximately **250 to 500 popular foreign TV series** released between 2010 and 2025 to ensure the analysis is relevant to modern streaming trends. The characteristics of the data will include:

- **Categorical Features:** Genre (e.g., Drama, Sci-Fi), original network or streaming platform, and series status (ongoing vs. cancelled).
- **Numerical Features:** Average IMDb rating, total number of votes, and normalized Google search interest indices.
- **Target Variable:** I will investigate if early-season search spikes can statistically predict the total number of seasons a show will have before it ends.